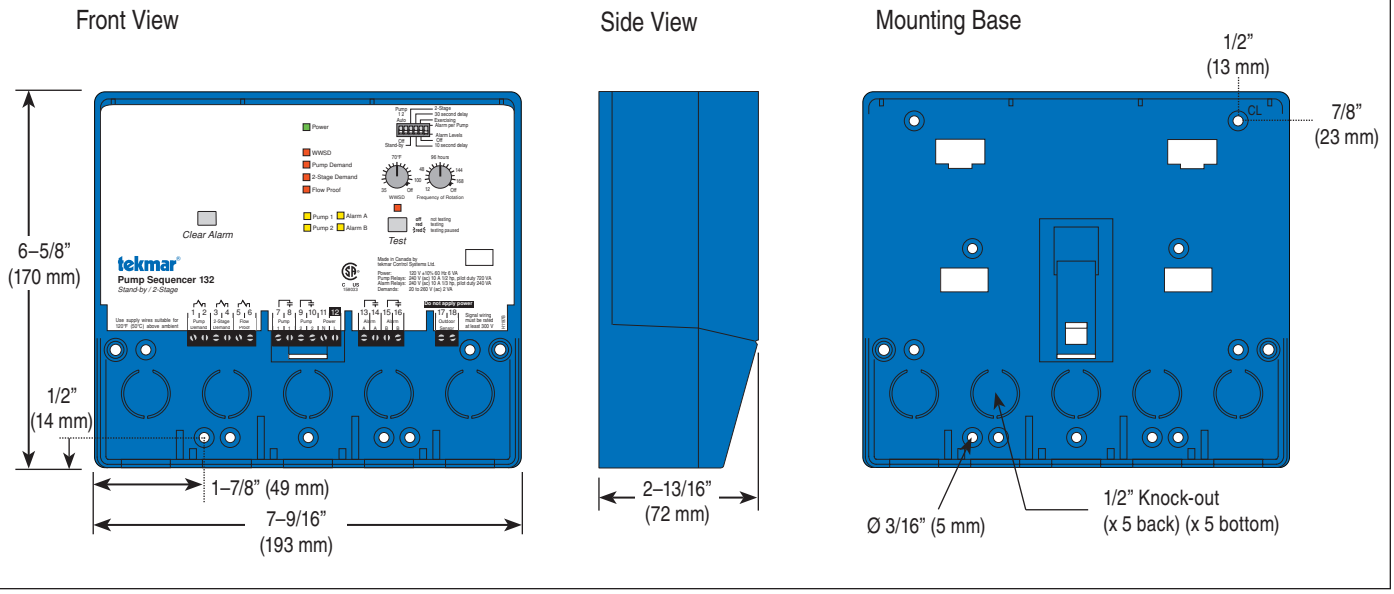
tekmar® Submittal
Pump Sequencer 132

Multi-Staging

C 132
01/11
Replaces: New

Job _____ Designer _____ Contact _____

The Pump Sequencer 132 is designed to operate 2 pumps by providing stand by or staging operation. It can be used in a wide range of applications, including residential installations using a backup system pump to commercial applications that require energizing a second stage for increased flow or head. The main function of the 132 is to provide lead lag capability to a standby pump setup, or to provide staging capability for a dual pump installation. The 132 can energize up to 2 pumps, as well as provide alert contacts that can be used in the event of a failure.



Specifications

Pump Sequencer 132 *Stand-by / 2-Stage*	
Literature	D132, A132, D001
Control	Microprocessor control. This is not a safety (limit) control
Packaged weight	2.9 lb. (1320 g)
Dimensions	6-5/8" H x 7-9/16" W x 2-13/16" D (170 x 193 x 72 mm)
Enclosure	Blue PVC plastic, NEMA type 1
Approvals	CSA C US, meets class B: ICES & FCC Part 15
Ambient conditions	Indoor use only, 32 to 120°F (0 to 50°C), RH ≤90% Non-condensing
Power supply	120 V ±10%, 50/60 Hz, 6 VA
Pump relays	240 V (ac) 10 A 1/2 hp
Alert relays	240 V (ac) 10 A 1/3 hp
Demands	20 to 260 V (ac) 2 VA
Sensors	NTC thermistor, 10 kΩ @ 77°F (25°C ±0.2°C) ß=3892
–Optional	Outdoor Sensor 070
Warranty	Limited 3 Year (See D132 for full warranty)

Energy Saving Features

- Warm Weather Shut Down

Additional Features

- Equal Run Time Rotation
- Lead lag operation
- 2 Stage capability
- Exercising
- Alert per pump or alert levels
- Adjustable flow proof delay
- Test sequence
- CSA C US certified
  (approved to applicable UL standards)

SPECIAL REQUIREMENTS

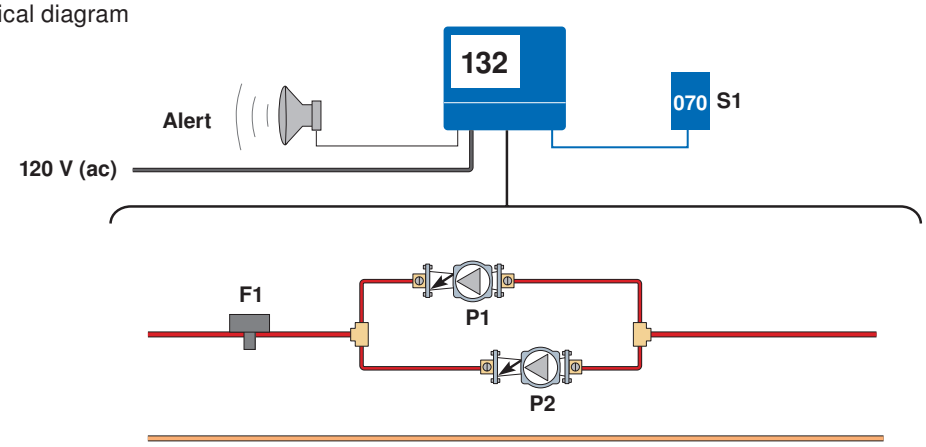
N / A

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Sample Application Drawing

Below is a sample application drawing for this product. This application may include other tekmar products that are required for installation. More sample applications can be found at www.tekmarcontrols.com.

Sample Mechanical diagram

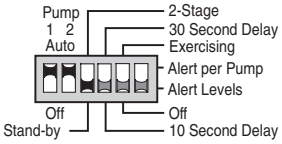


Sample Electrical diagram

Legend:

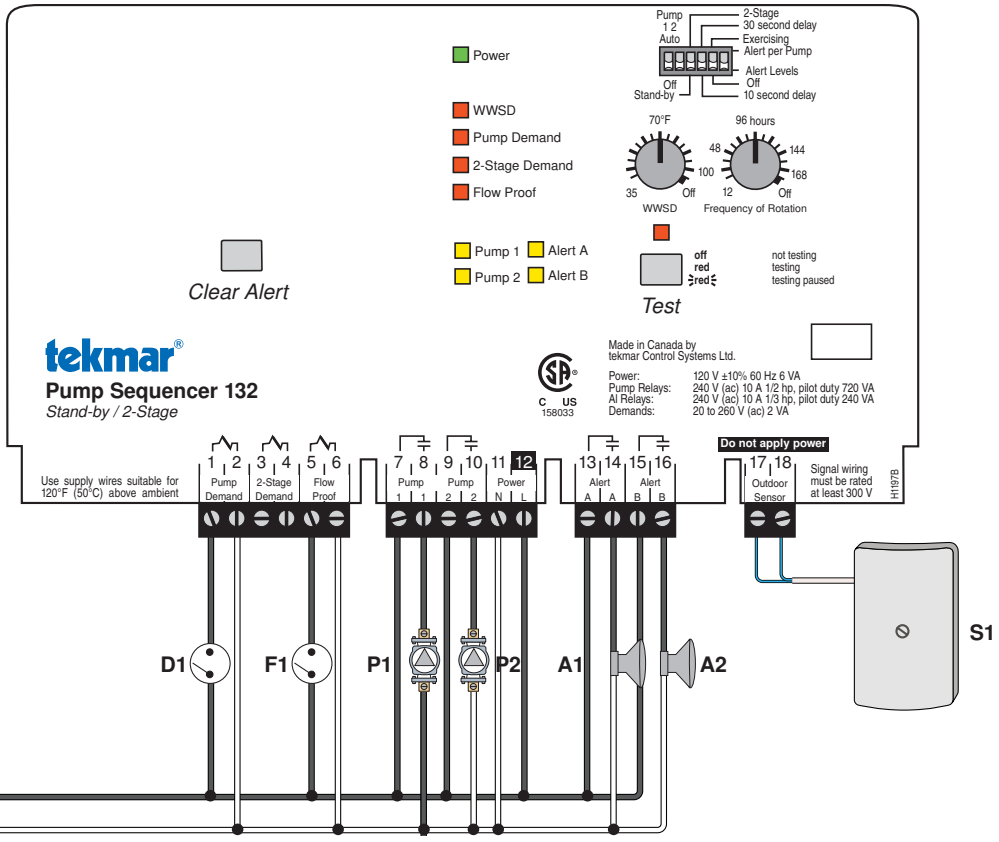
D1 = Pump Demand
F1 = Flow Switch
P1 = Pump 1
P2 = Pump 2
A1 = Alert 1
A2 = Alert 2
S1 = Outdoor Sensor

DIP Setting



= Required

= Optional



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tekmar® - Data Brochure

Pump Sequencer 132

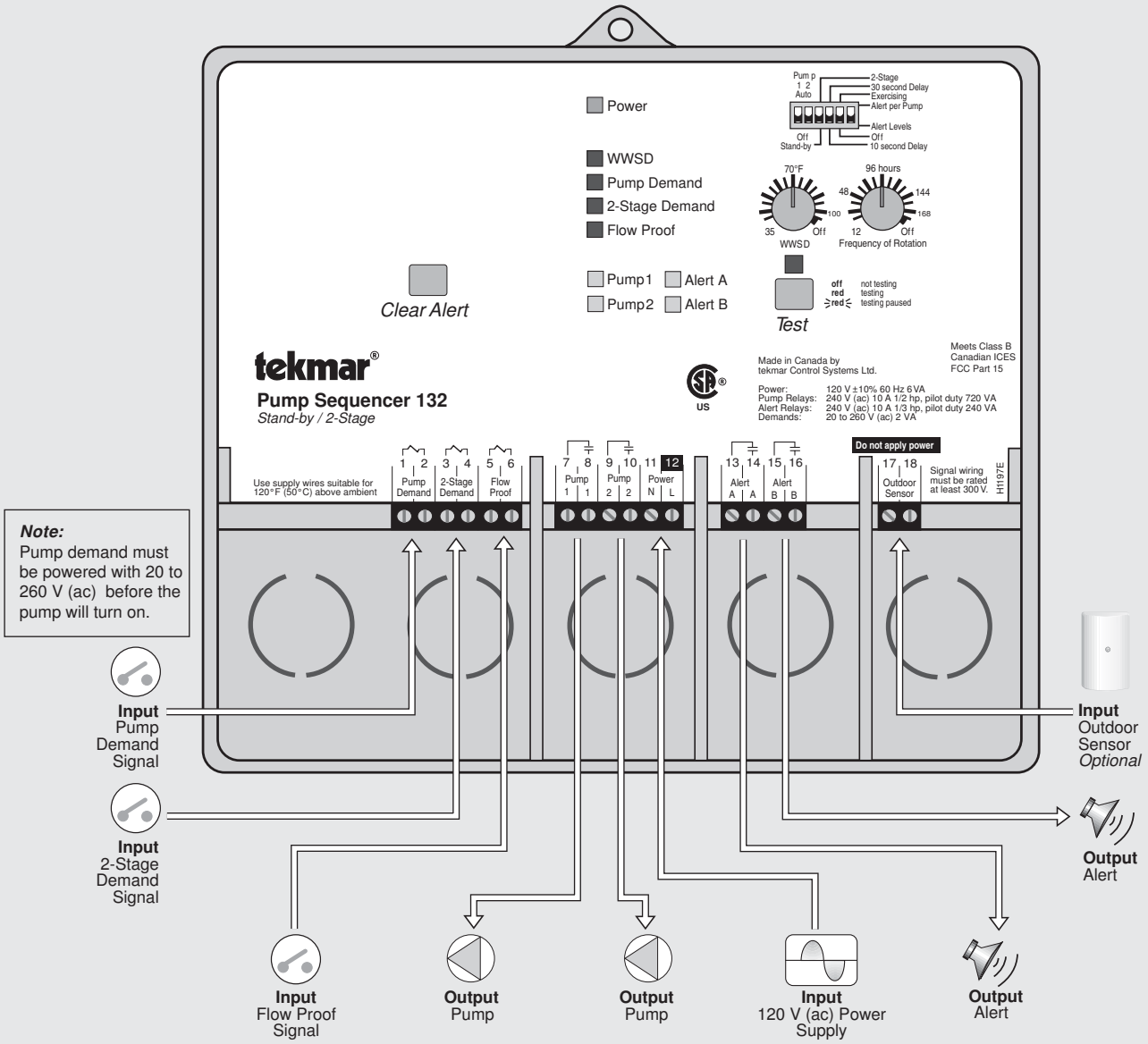
D 132

04/10

The Pump Sequencer 132 is designed to provide pump control for either stand-by or 2-stage operation. In stand-by mode, the 132 automatically switches over from the lead pump to the stand-by pump during a pump failure. In 2-stage mode, the 132 turns on the second stage pump to meet additional flow requirements.

Additional functions include:

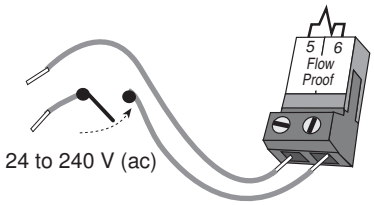
- Warm Weather Shut Down (WWSD)
- Exercising
- Equal Run Time Rotation
- Alert per Pump or Alert Levels
- Adjustable flow proof delay
- Test sequence to ensure proper component operation
- 120 V (ac) power supply
- CSA C US certified (approved to applicable UL standards)



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FLOW PROOF

A flow proof signal is required at all times during pump operation. A flow proof is generated by applying a voltage between 24 and 240 V (ac) across the *Flow Proof* terminals (5 and 6). Once voltage is applied, the *Flow Proof* light is turned on. Once a pump contact is turned on, a flow proof signal must be present before the flow proof delay has expired. A flow proof can come from a flow switch, pressure differential switch, current sensing switch, or power sensing switch.



FLOW PROOF DELAY *(30 second Delay / 10 second Delay)*

The 132 allows a time delay for detecting a flow proof signal once a pump contact is turned on. The amount of time is adjustable from 10 to 30 seconds through a DIP switch. If a flow proof signal is not present within the selected time, the control turns off the first pump contact and turns on the second pump contact. An alert contact is activated to indicate the failure.

FLOW PROOF TEST

The 132 has a flow proof test in order to determine if the flow proof device has failed. Once the pump contacts are turned off, a flow proof signal should not be present. If a flow proof signal is still present after 4 minutes, the control activates an alert contact and displays an error message. Refer to the *Error Messages* section at the back of this brochure.

ROTATION

The 132 has a function which automatically changes the operating sequence of the pumps based on *Equal Run Time Rotation*. *Equal Run Time Rotation* is based on pump running hours and allows for equal usage of both pumps. The 132 uses the *Frequency of Rotation* dial to set the rotation of the pumps.

The control rotates the operating sequence of the pumps when the pumps are off. In a constant circulation system where the lead pump runs continuously, the control waits for up to 12 hours to rotate the operating sequence of the pumps.

Note: *The* *Equal Run Time Rotation* function is reset by pressing the ***Clear Alert*** button.

EXERCISING

The 132 has a built-in pump exercising function. This function is only operational if the *Exercising / Off* DIP switch is set to Exercising. If a pump has not been operated at least once every three days, the control turns on the output for 10 seconds. This minimizes the possibility of a pump seizing during a long period of inactivity.

Note: The exercising function does not work if power to the control or pumps is off.

WARM WEATHER SHUT DOWN (WWSD)

When the 132 is used as a stand alone control, the pumps can be operated based on outdoor air temperature. The WWSD feature is only operational when an outdoor sensor is installed and the *WWSD* dial is not set to Off. When the outdoor air temperature rises above the WWSD setting, the 132 turns on the *WWSD* light. When the control is in Warm Weather Shut Down, the demand LED's are displayed if there is a demand. However, the control does not operate the pumps to satisfy these demands. If the 132 receives a pump demand from a reset control, the outdoor sensor should not be installed and the WWSD dial must be set to *Off*.

Section B: Pump Operation

Section B1
Stand-by

Section B2
2-Stage

Section B1: Stand-by

PUMP OPERATION

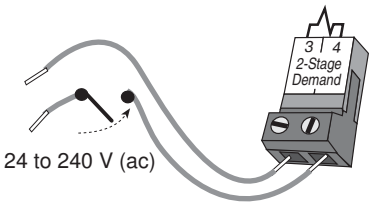
The lead pump contact closes whenever there is a pump demand and the 132 is not in WWSD. If a flow proof is not present after the flow proof delay has expired, the control turns off the lead pump contact and turns on the stand-by pump contact. An alert contact is activated to indicate the failure. The stand-by pump contact also turns off if a flow proof is not obtained once the flow proof delay has expired.

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Section B2: 2-Stage

2-STAGE DEMAND

A 2-stage demand is required in order for the 132 to provide additional flow. A 2-stage demand is generated by applying a voltage between 24 and 240 V (ac) across the *2-Stage Demand* terminals (3 and 4). Once voltage is applied, the *2-Stage Demand* light is turned on. A 2-stage demand can come from an additional pressure differential switch, a temperature differential device, or a setpoint control based on an outdoor air temperature.



Note: The 2-stage demand device should provide an appropriate differential to prevent the second stage pump from short cycling.

PUMP OPERATION

The first stage pump contact closes whenever there is a pump demand and the 132 is not in WWSD. The second stage pump contact closes whenever there is a pump demand, flow proof, 2-stage demand, and the 132 is not in WWSD.

If flow is not established by the first stage pump, the 132 turns off the first stage pump contact and turns on the second stage pump contact. An alert contact is activated to indicate the failure. The second stage pump contact also turns off if a flow proof is not obtained once the flow proof delay has expired.

Section C: Alert Operation

Section C1
Alert Levels

Section C2
Alert per Pump

There are two alert contacts (*Alert A and Alert B*) on the Pump Sequencer 132. These contacts are used to indicate either pump or control failures. The alert contacts have two modes of operation. The mode of operation for the alert contacts is selected using the *Alert per Pump / Alert Levels* DIP switch. When an alert contact is activated, refer to the *Error Messages* section of this brochure to determine the cause of the alert signal. To clear an alert, press the ***Clear Alert*** button.

Section C1: Alert Levels

When operating in the *Alert Levels* mode, the alert contacts are used to indicate the level of failure in the system.

ALERT A (Non-Critical)

The *Alert A* contact closes when a non-critical failure occurs. Non-critical failures include: a single pump failure, a control error, an outdoor sensor failure, or short cycling of the 2-stage demand. With these failures, it is still possible to establish flow in the system.

ALERT B (Critical)

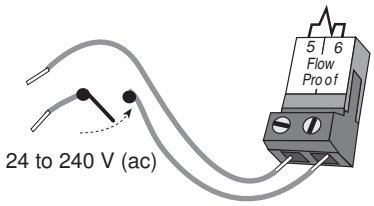
The *Alert B* contact closes when a critical failure occurs. Critical failures include: the failure of both pumps, or a failure of the flow proof device. With these failures, the control is no longer able to operate the pumps, and it is not possible to provide flow.

Section C2: Alert per Pump

When operating in *Alert per Pump* mode, the alert contacts are related to the pump contacts on the control. The *Alert A* contact closes anytime *Pump 1* fails, and the *Alert B* contact closes anytime *Pump 2* fails. The *Alert A* contact also closes to indicate an outdoor sensor failure, a control error, a flow proof device failure, or if the 2-stage demand is short cycling.

Flow Proof

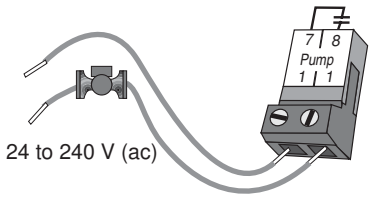
To generate a flow proof, a voltage between 24 V (ac) and 240 V (ac) must be applied across the *Flow Proof* terminals (5 and 6).



Output Connections

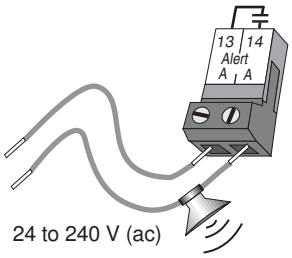
Pump Contacts

The *Pump 1* and *Pump 2* terminals (7, 8 and 9, 10) are isolated outputs in the 132. There is no power available on these terminals from the control. These terminals are to be used as a switch to either make or break power to the pump circuit. Since these are isolated contacts, they may switch a voltage between 24 V (ac) and 240 V (ac).



Alert Contacts

The *Alert A* and *Alert B* terminals (13, 14 and 15, 16) are isolated outputs in the 132. There is no power available on these terminals from the control. These terminals are to be used as a switch to either make or break power to the alert circuit. Since these are isolated contacts, they may switch a voltage between 24 V (ac) and 240 V (ac).



Outdoor Sensor Connection

Do not apply power to these terminals as this will damage the control.

Outdoor Sensor (Optional)

Connect the two wires from the Outdoor Sensor 070 to the *Outdoor Sensor* terminals (17 and 18). The outdoor sensor is used by the 132 to measure the outdoor air temperature.

Note: If an outdoor sensor is not installed, the *WWSD* dial must be set to Off.



STEP FIVE — TESTING THE WIRING

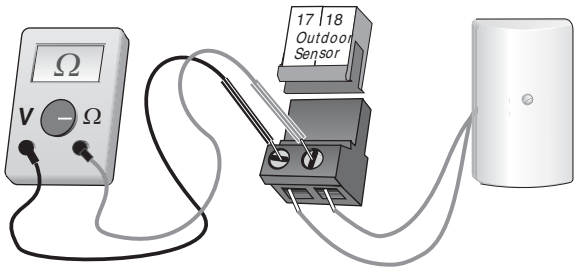
Each terminal block ***must be unplugged*** from its header on the control before power is applied for testing. To remove a terminal block, pull it straight down from the control.

The following tests are to be performed using standard testing practices and procedures, and should only be carried out by properly trained and experienced persons.

A good quality electrical test meter, capable of reading from at least 0 - 300 V (ac) and at least 0 - 2,000,000 Ohms, is essential to properly test the wiring and sensor.

Test The Sensor

In order to test the sensor, the actual temperature at the sensor location must be measured. A good quality digital thermometer with a surface temperature probe is recommended for ease of use and accuracy. Where a digital thermometer is not available, a spare sensor can be placed alongside the one to be tested and the readings compared. Test the sensor according to the instructions in the Data Brochure D 070.



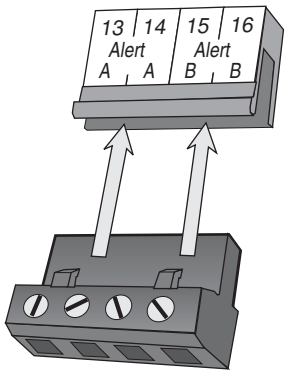
Connecting The Control

Make sure all power to the devices and terminal blocks is off, and remove any remaining jumpers from the terminals.

Reconnect the terminal blocks to the control by carefully aligning them with their respective headers on the control and then pushing the terminal blocks into the headers. The terminal blocks should snap firmly into place.

Install the supplied safety dividers between the unpowered sensor input and the powered 120 V (ac) or 24 V (ac) wiring chambers.

Apply power to the control. The operation of the control on power up is described in the *Sequence of Operation* section of this brochure.

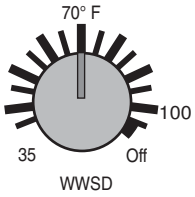


Settings

WARM WEATHER SHUT DOWN (WWSD)

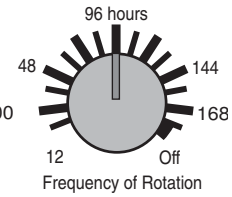
The *WWSD* dial is only operational when an outdoor sensor is installed. The WWSD dial can be set from 35 to 100°F or Off (2 to 38°C or Off). Refer to section A.

Note: If an outdoor sensor is not installed, the *WWSD* dial must be set to *Off*.



FREQUENCY OF ROTATION

The *Frequency of Rotation* dial can be set from 12 to 168 hours or Off. Refer to section A.



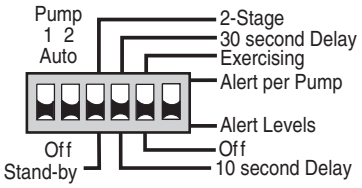
DIP SWITCH SETTINGS

Pump 1 Auto / Off

The *Pump 1 Auto / Off* DIP switch is used to select pump 1 operation. If the DIP switch is set to *Pump 1 Auto*, refer to section B.

Pump 2 Auto / Off

The *Pump 2 Auto / Off* DIP switch is used to select pump 2 operation. If the DIP switch is set to *Pump 2 Auto*, refer to section B.



2-Stage / Stand-by

The 2-Stage / Stand-by DIP switch is used to select which mode of pump control is required. If 2-Stage is selected, refer to section B2, and if Stand-by is selected, refer to section B1.

30 second Delay / 10 second Delay

The *30 second Delay / 10 second Delay* DIP switch is used to select what time delay is required to prove flow once a pump contact is turned on. Refer to section A.

Exercising / Off

The *Exercising / Off* DIP switch is used to select exercising. If the DIP switch is set to *Exercising*, refer to section A.

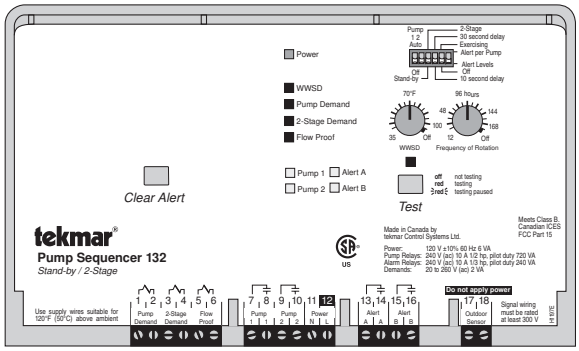
Alert per Pump / Alert Levels

The *Alert per Pump / Alert Levels* DIP switch is used to select which mode of alert operation is required. Refer to section C.

Technical Data

Pump Sequencer 132 *Stand-by / 2-Stage*

Literature	—	D 132, A 132's, D 001.
Control	—	Microprocessor control; This is **not a safety (limit) control**.
Packaged weight	—	2.9 lb. (1320 g), Enclosure A, blue PVC plastic
Dimensions	—	6-5/8" H x 7-9/16" W x 2-13/16" D (170 x 193 x 72 mm)
Approvals	—	CSA US, meets ICES & FCC regulations for EMI/RFI.
Ambient conditions	—	Indoor use only, 32 to 120°F (0 to 50°C), < 90% RH non-condensing.
Power supply	—	120 V (ac) ±10% 60 Hz 6 VA
Pump Relays	—	240 V (ac) 10 A 1/2 hp, pilot duty 720 VA
Alert Relays	—	240 V (ac) 10 A 1/3 hp, pilot duty 240 VA
Demands	—	20 to 260 V (ac) 2 VA
Sensors	—	NTC thermistor, 10 kΩ @ 77°F (25°C ±0.2°C) ß=3892
Optional	—	Outdoor Sensor 070



The installer must ensure that this control and its wiring are isolated and/or shielded from strong sources of electromagnetic noise. Conversely, this Class B digital apparatus complies with Part 15 of the FCC Rules and meets all requirements of the Canadian Interference-Causing Equipment Regulations. However, if this control does cause harmful interference to radio or television reception, which is determined by turning the control off and on, the user is encouraged to try to correct the interference by re-orientating or relocating the receiving antenna, relocating the receiver with respect to this control, and/or connecting the control to a different circuit from that to which the receiver is connected.

Cet appareil numérique de la classe B respecte toutes les exigences du Règlement sur le matériel brouilleur du Canada.

Caution The nonmetallic enclosure does not provide grounding between conduit connections. Use grounding type bushings and jumper wires.

Attention Un boîtier nonmétallique n´assure pas la continuité électrique des conduits. Utiliser des manchons ou des fils de accord spécialement conçus pour la mise á la terre.

Limited Warranty and Product Return Procedure

Limited Warranty ***The liability of tekmar under this warranty is limited. The Purchaser, by taking receipt of any tekmar product ("Product"), acknowledges the terms of the Limited Warranty in effect at the time of such Product sale and acknowledges that it has read and understands same.***

The tekmar Limited Warranty to the Purchaser on the Products sold hereunder is a manufacturer's pass-through warranty which the Purchaser is authorized to pass through to its customers. Under the Limited Warranty, each tekmar Product is warranted against defects in workmanship and materials if the Product is installed and used in compliance with tekmar's instructions, ordinary wear and tear excepted. The pass-through warranty period is for a period of twenty-four (24) months from the production date if the Product is not installed during that period, or twelve (12) months from the documented date of installation if installed within twenty-four (24) months from the production date.

The liability of tekmar under the Limited Warranty shall be limited to, at tekmar's sole discretion: the cost of parts and labor provided by tekmar to repair defects in materials and/or workmanship of the defective product; or to the exchange of the defective product for a warranty replacement product; or to the granting of credit limited to the original cost of the defective product, and such repair, exchange or credit shall be the sole remedy available from tekmar, and, without limiting the foregoing in any way, tekmar is not responsible, in contract, tort or strict product liability, for any other losses, costs, expenses, inconveniences, or damages, whether direct, indirect, special, secondary, incidental or consequential, arising from ownership or use of the product, or from defects in workmanship or materials, including any liability for fundamental breach of contract.

The pass-through Limited Warranty applies only to those defective Products returned to tekmar during the warranty period. This Limited Warranty does not cover the cost of the parts or labor to remove or transport the defective Product, or to reinstall the repaired or replacement Product, all such costs and expenses being subject to Purchaser's agreement and warranty with its customers.

Any representations or warranties about the Products made by Purchaser to its customers which are different from or in excess of the tekmar Limited Warranty are the Purchaser's sole responsibility and obligation. Purchaser shall indemnify and hold tekmar harmless from and against any and all claims, liabilities and damages of any kind or nature which arise out of or are related to any such representations or warranties by Purchaser to its customers.

The pass-through Limited Warranty does not apply if the returned Product has been damaged by negligence by persons other than tekmar, accident, fire, Act of God, abuse or misuse; or has been damaged by modifications, alterations or attachments made subsequent to purchase which have not been authorized by tekmar; or if the Product was not installed in compliance with tekmar's instructions and/or the local codes and ordinances; or if due to defective installation of the Product; or if the Product was not used in compliance with tekmar's instructions.

THIS WARRANTY IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, WHICH THE GOVERNING LAW ALLOWS PARTIES TO CONTRACTUALLY EXCLUDE, INCLUDING, WITHOUT LIMITATION, IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, DURABILITY OR DESCRIPTION OF THE PRODUCT, ITS NON-INFRINGEMENT OF ANY RELEVANT PATENTS OR TRADEMARKS, AND ITS COMPLIANCE WITH OR NON-VIOLATION OF ANY APPLICABLE ENVIRONMENTAL, HEALTH OR SAFETY LEGISLATION; THE TERM OF ANY OTHER WARRANTY NOT HEREBY CONTRACTUALLY EXCLUDED IS LIMITED SUCH THAT IT SHALL NOT EXTEND BEYOND TWENTY-FOUR (24) MONTHS FROM THE PRODUCTION DATE, TO THE EXTENT THAT SUCH LIMITATION IS ALLOWED BY THE GOVERNING LAW.

Product Warranty Return Procedure All Products that are believed to have defects in workmanship or materials must be returned, together with a written description of the defect, to the tekmar Representative assigned to the territory in which such Product is located. If tekmar receives an inquiry from someone other than a tekmar Representative, including an inquiry from Purchaser (if not a tekmar Representative) or Purchaser's customers, regarding a potential warranty claim, tekmar's sole obligation shall be to provide the address and other contact information regarding the appropriate Representative.

tekmar® Control Systems

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